

Project FORESIGHT

**AI-Powered Demand &
Inventory Intelligence Platform**

Zidio Internship Project

Developed By

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Project Duration:

One month

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Project Overview:

Project FORESIGHT – AI-Powered Demand & Inventory Intelligence Platform is a data-driven analytics solution designed to help businesses make smarter demand forecasting and inventory management decisions. The project combines Data Analytics, Machine Learning, Power BI, DAX, and interactive visualization to analyse historical sales and inventory data and generate actionable business insights.

The insights are presented through an interactive Power BI dashboard and Stream lit application, allowing users to monitor key performance indicators, analyse demand trends, identify high-risk products, and understand inventory requirements.

Machine learning techniques are used to forecast future product demand, while analytical calculations help determine inventory requirements and support better replenishment planning.

Objectives of the Project:

1. Demand Forecasting:

To analyse historical sales data and use machine learning techniques to forecast future demand at the product/SKU level.

2. Inventory Risk Identification:

To identify products that are at risk of stockouts, or overstocking based on demand and available inventory.

3. Inventory Optimization:

To support businesses in maintaining appropriate inventory levels and reducing excess or insufficient stock.

4. Reorder Recommendations:

To provide data-driven recommendations regarding when and how much inventory should be replenished.

5.Improve Supply Chain Planning

To provide predictive insights that can help businesses plan procurement and replenishment more effectively and reduce the impact of demand uncertainty.

Architecture Overview:

Project FORESIGHT – AI-Powered Demand & Inventory Intelligence Platform follows a structured data-to-decision architecture that integrates data processing, machine learning, inventory analytics, and business intelligence into a single workflow.

The architecture consists of multiple layers, where raw sales and inventory data is transformed into meaningful insights, demand forecasts, risk indicators, and inventory recommendations.

Architecture Flow:

Data Sources



Data Ingestion



Data Cleaning & Preprocessing



Exploratory Data Analysis & Feature Engineering



Machine Learning – Demand Forecasting



Inventory Intelligence & Risk Analysis



Business Intelligence Layer



Power BI and Streamlit Dashboard



Insights and Inventory Recommendations

Architecture Components:

1. Data Sources

The system uses historical business data containing information such as sales, products/SKUs, inventory levels, dates, demand, pricing, and other relevant operational attributes.

2. Data Ingestion

The collected data is imported from sources such as Excel or CSV files and prepared for further analysis.

3. Data Preprocessing

Python-based data processing techniques are used to handle missing values, duplicates

4. Stream lit Application:

A Stream lit-based interface provides an interactive analytical environment where users can explore demand forecasts, inventory status, risk indicators, and recommendations.

5. Decision and Recommendation Layer

The final outputs help users understand what products may require attention, where inventory risks exist, and how demand trends can influence replenishment decisions.

6. Business Intelligence Layer

The processed results are connected to Power BI, where DAX measures, KPIs, charts, slicers, and interactive reports are used to present business insights.

7. Stream lit Application

A Stream lit-based interface provides an interactive analytical environment where users can explore demand forecasts, inventory status, risk indicators, and recommendations.

Project Execution Timeline:

The development of Project FORESIGHT – AI-Powered Demand & Inventory Intelligence Platform was carried out in a structured sequence, starting from requirement analysis and data collection and progressing through data preparation, machine learning, dashboard development, testing, and documentation.

Execution Timeline:

Week 1 (24 April – 30 April 2026)

- Requirement gathering and project planning
- Dataset collection and preprocessing
- Exploratory Data Analysis

Week 2 (1 May – 7 May 2026)

- Feature engineering
- Demand pattern and inventory trend analysis
- Machine learning model preparation and implementation

Week 3 (8 May – 15 May 2026)

- Demand forecasting model development
- Inventory risk analysis and optimization
- Reorder point and safety stock calculation
- Integration of forecasting and inventory intelligence

Week 4 (16 May – 24 May 2026)

- Power BI dashboard development
- Streamlit dashboard development
- DAX measures, KPIs, and interactive visualizations
- Testing, validation, and final integration
- Final project documentation and report preparation

Streamlit Dashboard:

The Streamlit Dashboard is an interactive web-based component of Project FORESIGHT developed using Python and Streamlit. It provides a simple interface for users to explore demand forecasts, inventory performance, stock risks, and business insights without directly working with Python code.

Key Features:

- **Interactive Dashboard:** Displays important project results through an easy-to-use web interface.
- **KPI Cards:** Shows key metrics such as Total Sales, Total Inventory, Total Demand, Forecasted Demand, and Inventory Value.
- **Demand Forecasting:** Presents predicted demand and allows users to analyse future demand patterns.
- **Inventory Analysis:** Displays current inventory levels and identifies products requiring attention.
- **Stockout Analysis:** Highlights products with insufficient or zero inventory.
- **Dashboard Workflow**
- Data → Python Processing → ML Forecasting → Inventory Analysis → Streamlit → Interactive Insights

Technical Highlights:

- **Python-Based Data Processing:** Used Python libraries such as Pandas and NumPy for data cleaning, transformation, preprocessing, and analysis.
- **Exploratory Data Analysis (EDA):** Analysed sales, demand, inventory, and product-level trends to identify patterns, anomalies, and business insights.
- **Feature Engineering:** Created meaningful features from historical sales and inventory data to improve demand analysis and machine learning performance.
- **Machine Learning:** Implemented machine learning techniques for demand forecasting and predictive inventory analysis.

- **Demand Forecasting:** Developed forecasting capabilities to estimate future product demand using historical demand patterns.
- **DAX Measures:** Created DAX calculations for important metrics such as Total Sales, Average Demand, Inventory Value, Stockout Risk, and Inventory KPIs.
- **Interactive Visualization:** Used charts, slicers, filters, and KPI cards to enable users to explore product and inventory performance dynamically.
- **End-to-End Analytics Pipeline:** Integrated data preprocessing → EDA → feature engineering → ML forecasting → inventory intelligence → Power BI/Streamlit visualization into a unified analytical solution.
- **Business Decision Support:** Converted analytical and predictive outputs into actionable insights to support demand planning, inventory management, and replenishment decisions.

DAX Measures and Formulas:

The following DAX formulas were used in Project FORESIGHT for sales analysis, demand forecasting, inventory monitoring, risk identification, and replenishment planning.

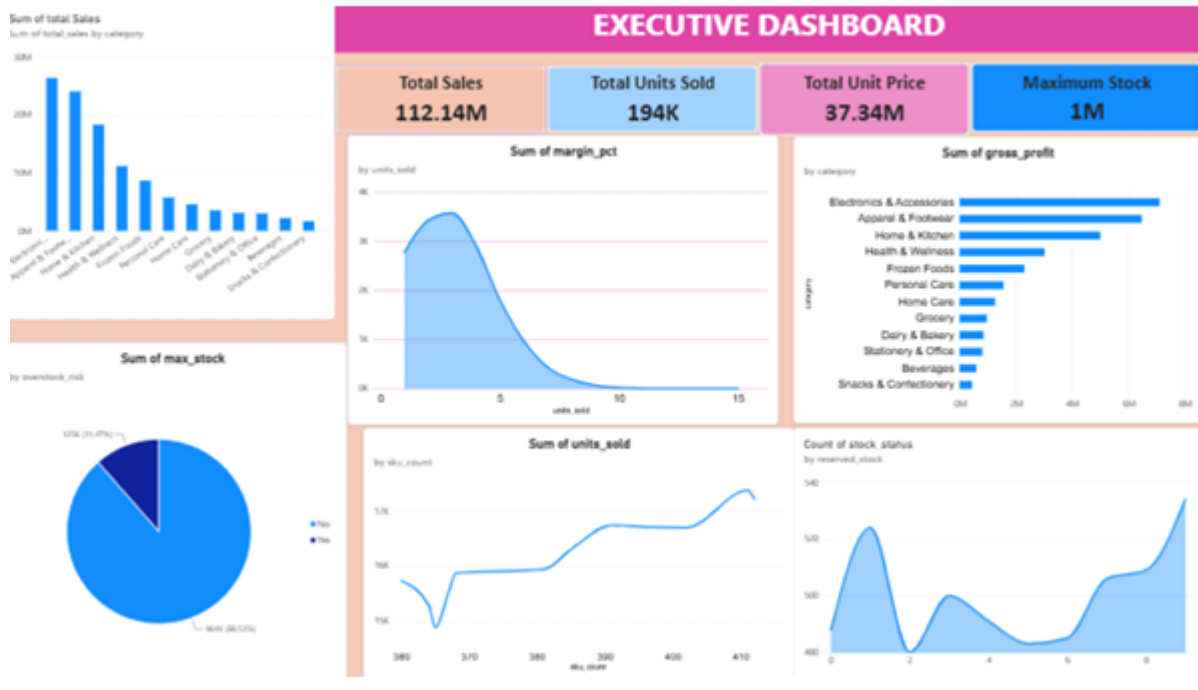
- **Total Sales:** Total Sales = SUM ('Inventory Data'[Sales])
- **Total Quantity Sold:** Total Quantity Sold = SUM ('Inventory Data'[Quantity Sold])
- **Total Inventory:** Total Inventory = SUM ('Inventory Data'[Inventory Quantity])
- **Total Demand:** Total Demand = SUM ('Inventory Data'[Demand])
- **Average Demand:** Average Demand = AVERAGE ('Inventory Data'[Demand])
- **Forecasted Demand:** Forecasted Demand = SUM ('Inventory Data'[Forecasted Demand])
- **Inventory Value:** Inventory Value = SUMX ('Inventory Data', 'Inventory Data'[Inventory Quantity] * 'Inventory Data'[Unit Price])
- **Stockout Products:** Stockout Products = CALCULATE (DISTINCTCOUNT ('Inventory Data'[Product ID]), 'Inventory Data'[Inventory Quantity] = 0)
- **Stockout Products:** Stockout Products = CALCULATE (DISTINCTCOUNT ('Inventory Data'[Product ID]), 'Inventory Data'[Inventory Quantity] = 0)
- **Low Stock Products:** Low Stock Products = CALCULATE (DISTINCTCOUNT ('Inventory Data'[Product ID]), 'Inventory Data'[Inventory Quantity] < 'Inventory Data'[Reorder Point])
- **Overstock Products:** Overstock Products = CALCULATE (DISTINCTCOUNT ('Inventory Data'[Product ID]), 'Inventory Data'[Inventory Quantity] > 'Inventory Data'[Maximum Stock Level])
- **Reorder Quantity:** Reorder Quantity = SUMX ('Inventory Data', MAX (0, 'Inventory Data'[Reorder Point] - 'Inventory Data'[Inventory Quantity]))

Technologies Used

- Python – Data processing, analysis, feature engineering, and machine learning.
- Pandas – Data cleaning, transformation, and dataset manipulation.
- NumPy – Numerical calculations and data processing.
- Scikit-learn – Machine learning model development and evaluation.
- Matplotlib & Seaborn – Data visualization and exploratory data analysis.
- Jupyter Notebook – Model development, analysis, and experimentation.
- Microsoft Excel – Data storage, preparation, and initial analysis.
- Power BI – Interactive dashboards, KPIs, inventory analysis, and business intelligence.
- DAX – Creating calculated measures and inventory-related KPIs in Power BI.
- Power Query – Data transformation and preparation for Power BI.
- Streamlit – Development of the interactive web-based analytics dashboard.
- Git & GitHub – Version control and project documentation.
- Docker – Application containerization and deployment.

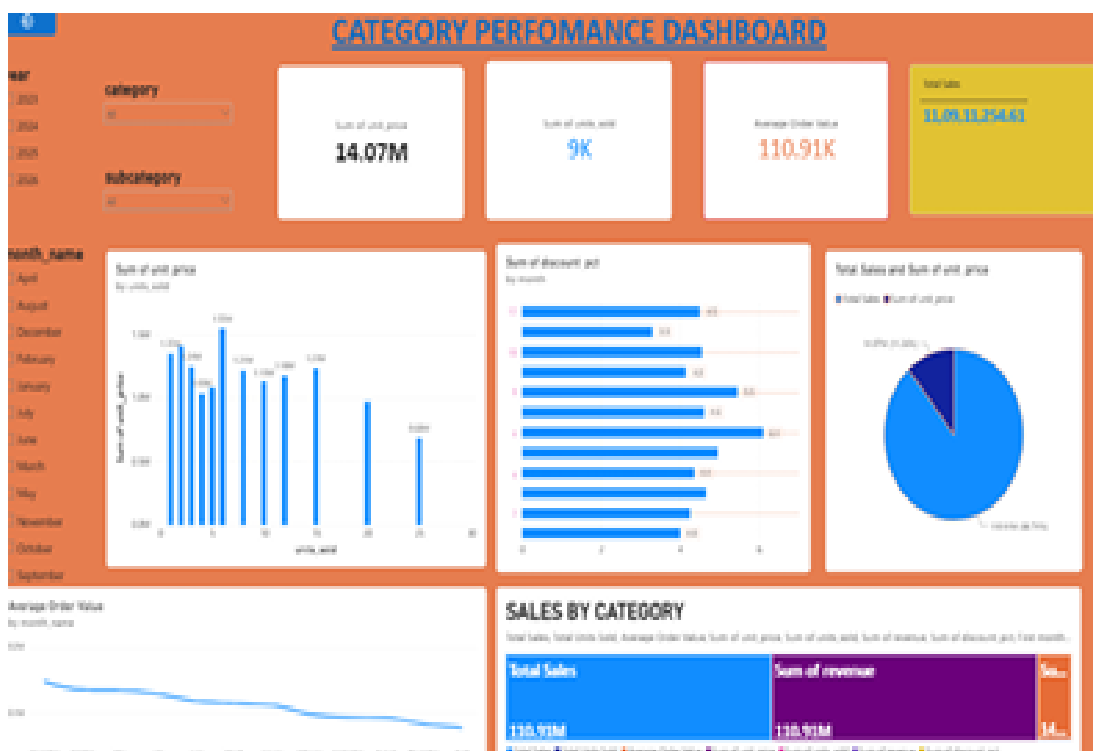
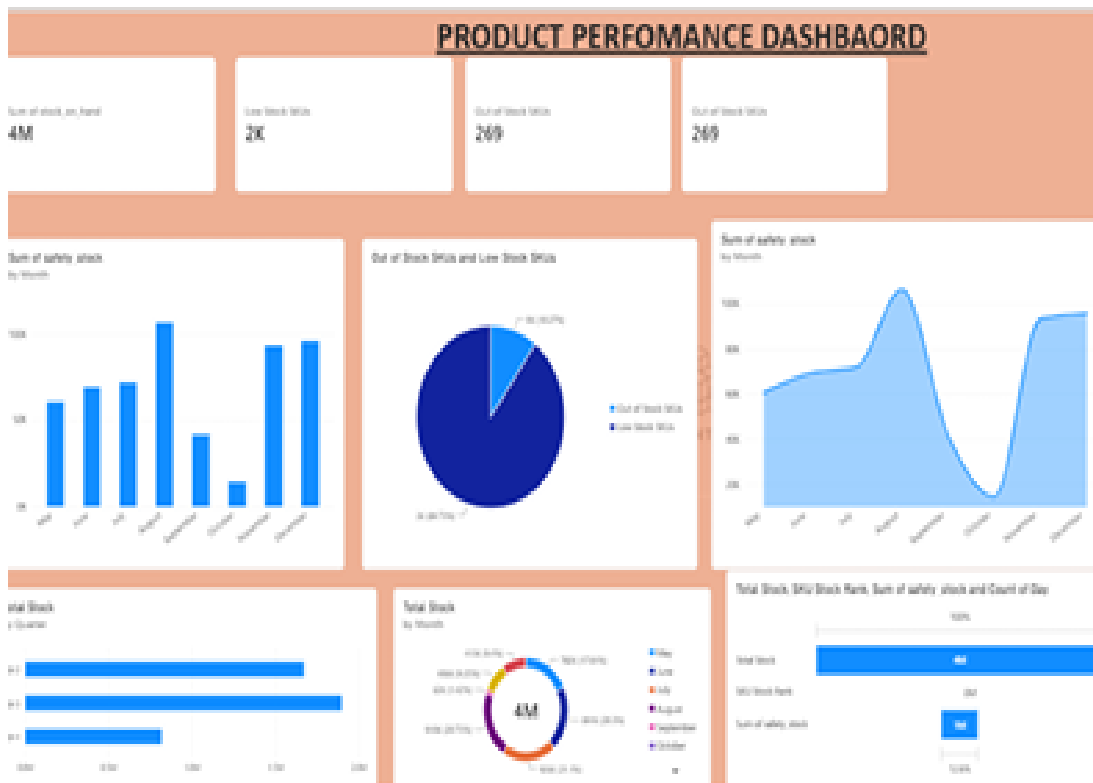
DASHBOARD VISUALS:

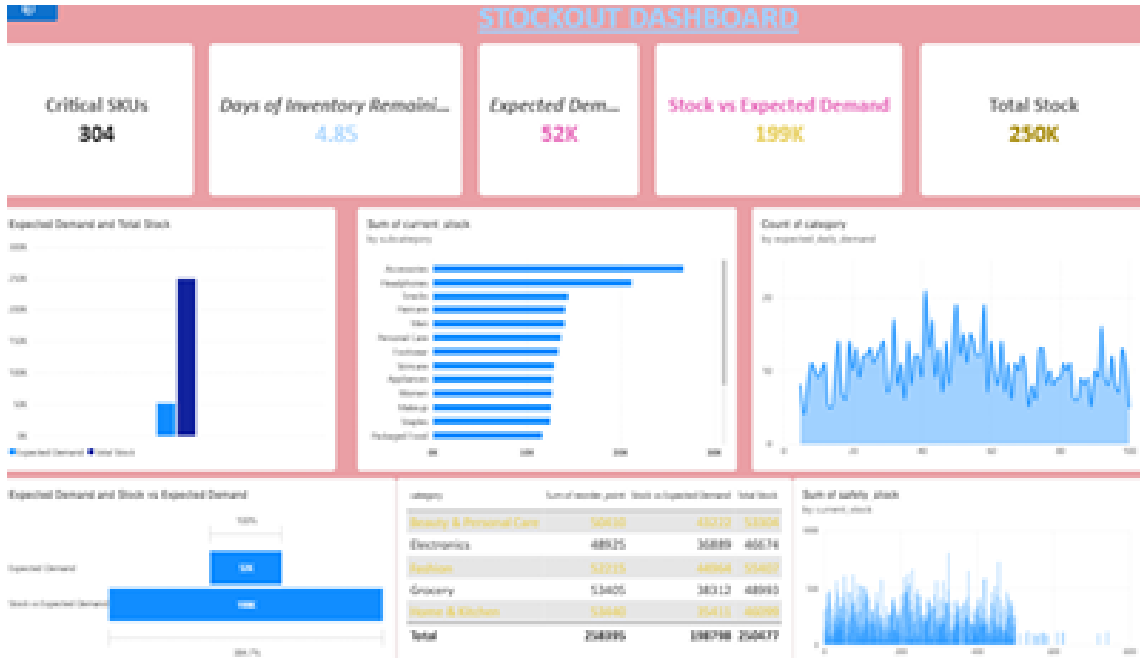
1.Executive Dashboard:



2.Sales Dashboard









Project Output:

The **FORESIGHT – AI-Powered Demand & Inventory Intelligence Platform** produces the following key outputs:

- **Demand Forecasting:** Predicts future product demand using historical sales and inventory data.
- **Inventory Analysis:** Provides an overview of current inventory levels and product availability.
- **Stockout Identification:** Identifies products with insufficient or zero inventory.
- **Overstock Detection:** Highlights products with excess inventory.
- **Reorder Recommendations:** Identifies products that require replenishment based on demand and inventory levels.
- **Safety Stock Analysis:** Helps determine the inventory level required to handle demand uncertainty.

- **Inventory KPIs:** Provides metrics such as total sales, total demand, inventory value, inventory turnover, and forecast accuracy.

Conclusion:

->The FORESIGHT project demonstrates how Machine Learning, Data Analytics, Power BI, DAX, and Stream lit can be integrated to build an intelligent demand and inventory management solution. The system transforms raw historical data into meaningful insights through data preprocessing, exploratory analysis, demand forecasting, and inventory risk analysis.

->The resulting dashboards provide a clear view of demand trends, inventory levels, stockout risks, overstock conditions, and replenishment requirements. By combining predictive forecasting with business intelligence, FORESIGHT supports more informed inventory planning and helps organizations respond to changing demand patterns.

->Overall, the project provides an end-to-end data-driven framework for demand forecasting and inventory intelligence, demonstrating the practical application of analytics and machine learning in supply chain and business decision-making

